

1)

p	q	r	$\neg q$	$p \wedge \neg q$	$(p \wedge \neg q) \rightarrow r$
T	T	T	F	F	T
T	T	F	F	F	T
T	F	T	T	T	T
T	F	F	T	T	F
F	T	T	F	F	T
F	T	F	F	F	T
F	F	T	T	F	T
F	F	F	T	F	T

2)

We have: c = 'The car is red', m = 'The car is a manual transmission', d = 'The car is a diesel engine', h = 'The car has high mileage'.

2(a)

$\neg c \rightarrow \neg h$: If the car isn't red, then the car does not have high mileage

$c \rightarrow (\neg m \vee \neg d)$: if the car is red, then the car is not a manual transmission or the car is not a diesel engine

$(c \wedge m) \rightarrow \neg d$: If the car is red and the car is a manual transmission, then the car is not a diesel engine

2(b)

negation of $\neg c \rightarrow \neg h$:

$$\neg(\neg c \rightarrow \neg h) \equiv \neg(c \vee \neg h) \equiv \neg c \wedge h$$

Negation of $c \rightarrow (\neg m \vee \neg d)$:

$$\neg(c \rightarrow (\neg m \vee \neg d)) \equiv \neg(\neg c \vee (\neg m \vee \neg d)) \equiv c \wedge \neg(\neg m \vee \neg d) \equiv c \wedge (m \wedge d) \equiv c \wedge m \wedge d$$

So answer is: $c \wedge m \wedge d$

3)

Statement: If the temperature is below freezing then the lake is frozen.

p = 'temperature is below freezing', q = 'lake is frozen'. So the statement is $p \rightarrow q$.

3(1)

Converse ($q \rightarrow p$): If the lake is frozen, then the temperature is below freezing.

Inverse ($\neg p \rightarrow \neg q$): If the temperature is not below freezing, then the lake is not frozen

contrapositive ($\neg q \rightarrow \neg p$): If the lake is not frozen, then the temperature is not below freezing.

3(2) Negation

Negation of $p \rightarrow q$ is $p \wedge \neg q$: The temperature is below freezing and the lake is not frozen

4)

$$4(1) (p \vee q) \wedge (\neg p \wedge \neg q) \equiv F$$

$$(p \vee q) \wedge (\neg p \wedge \neg q)$$

$$\equiv ((p \vee q) \wedge \neg p) \wedge \neg q$$

$$\equiv ((p \wedge \neg p) \vee (q \wedge \neg p)) \wedge \neg q$$

$$\equiv (F \vee (q \wedge \neg p)) \wedge \neg q$$

$$\equiv (q \wedge \neg p) \wedge \neg q$$

$$\equiv (q \wedge \neg q) \wedge \neg p$$

$$\equiv F \wedge \neg p$$

$$\equiv F$$

$$4(2) (p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$$

$$(p \rightarrow q) \wedge (p \rightarrow r)$$

$$\equiv (\neg p \vee q) \wedge (\neg p \vee r)$$

$$\equiv \neg p \vee (q \wedge r)$$

$$\equiv p \rightarrow (q \wedge r)$$

$$4(3) [(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r) \equiv T$$

$$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$$

$$\begin{aligned}
&\equiv [(\neg p \vee q) \wedge (\neg q \vee r)] \rightarrow (\neg p \vee r) \\
&\equiv \neg[(\neg p \vee q) \wedge (\neg q \vee r)] \vee (\neg p \vee r) \\
&\equiv (\neg(\neg p \vee q) \vee \neg(\neg q \vee r)) \vee (\neg p \vee r) \\
&\equiv ((p \wedge \neg q) \vee (q \wedge \neg r)) \vee (\neg p \vee r) \\
&\equiv (\neg p \vee r \vee (p \wedge \neg q)) \vee (q \wedge \neg r) \\
&\equiv ((\neg p \vee r \vee p) \wedge (\neg p \vee r \vee \neg q)) \vee (q \wedge \neg r) \\
&\equiv (T \wedge (\neg p \vee r \vee \neg q)) \vee (q \wedge \neg r) \\
&\equiv (\neg p \vee r \vee \neg q) \vee (q \wedge \neg r) \\
&\equiv ((\neg p \vee r \vee \neg q \vee q) \wedge (\neg p \vee r \vee \neg q \vee \neg r)) \\
&\equiv ((\neg p \vee r \vee T) \wedge (\neg p \vee (r \vee \neg r) \vee \neg q)) \\
&\equiv (T \wedge (\neg p \vee T \vee \neg q)) \\
&\equiv T
\end{aligned}$$

5)

So we have: $(\neg \text{tired} \wedge \neg \text{busy}) \rightarrow \text{go}$. Let $t = \text{tired}$, $b = \text{busy}$, $g = \text{go}$.

$$(\neg t \wedge \neg b) \rightarrow g$$

$$\equiv \neg(\neg t \wedge \neg b) \vee g$$

$$\equiv (t \vee b) \vee g$$

$$\equiv t \vee b \vee g$$

So finally it would be: $t \vee b \vee \text{go}$